**5** Two blocks A and B have masses 5 kg and 2 kg respectively. The blocks are joined together by a light rigid horizontal bar and slide across a horizontal surface. The resistances on the blocks A and B are modelled as 7 N and 1.75 N respectively. There are no other external horizontal forces.

**(a)** Calculate the acceleration of the system, according to the model. **[3]**

**(b)** In reality, the acceleration of the system is **not** constant. State an improvement that could be made to the model so that it will produce results which are closer to the real situation. **[1]**

**5(a)**

**5(b)**